LAB 03 | Chaotic Text Generator

Problem and solution. Generate sentences without reference outputs, compare structural validity, plausibility, novelty and lexical diversity, and decode POS tags from input text. We train count-based first-order HMMs with coarse or role-split hidden tags, plus a trigram baseline. Sampling is controlled by temperature; Viterbi uses smoothed log-probabilities. Data is split by document: 960 synthetic training sentences and 240 test sentences. Metrics use 500 generated sentences per model, seed 42.

1. Generation at temperature T = 1.0

Model	Valid pattern	Plausible S–V	Novel	Distinct-2
Plain HMM	16.2%	14.8%	99.8%	0.198
Role-split HMM	100.0%	60.6%	99.4%	0.148
Trigram	57.0%	76.0%	91.2%	0.105

The plain HMM only sees the preceding coarse POS tag: NOUN can stand for a subject, object or prepositional object, so transitions often form impossible sequences. Splitting DET/NOUN/ADJ into S, O and P roles preserves position in the sentence and gives 100% structural validity on this tiny grammar. The trigram captures local lexical context, improving subject–verb plausibility but sometimes copying training text.

2. Temperature selection and failures

Writing toy: T = 1.5 with the role-split HMM: more lexical variety (distinct-2 = 0.161 vs 0.148 at T = 1.0) while keeping 100% valid tag patterns. **Autocomplete:** T = 0.5 with the trigram: more predictable continuations and better structural validity (63.6% vs 44.2% at T = 2.5), though copied text must be flagged.
Meaning failure: “the foxes lose by the school .” has valid POS structure but a subject–verb pairing absent from training; role splitting does not model semantics. **Memorisation failure:** “the girls leave the park .” is copied verbatim from training by the low-temperature trigram. The plain HMM also shows structural failures because it merges distinct noun roles.

3. Synthetic versus real text

Dataset / model	Token acc.	Known	Unknown	Whole sentence
Synthetic / plain HMM	100.0%	—	—	100.0%
UD EWT / most-frequent tag	86.2%	90.8%	28.6%	31.9%
UD EWT / HMM+Viterbi	88.0%	92.9%	28.2%	40.4%

The UD EWT figures come from a fresh run of `python -m src.evaluate_real` (12,544 training / 2,077 test sentences; results/ewt_results.json). Real English has 17 POS tags, a much larger vocabulary, 10,359 distinct training tag sequences, and unknown words make up 7.5% of test tokens. The synthetic grammar has only 16 observed patterns and unambiguous context. The HMM gains only 1.8 points over the most-frequent-tag baseline on real tokens; unseen-word emissions and ambiguity (e.g. PROPN→NOUN, 281 errors) cause the rest. So 100% synthetic tagging and valid-pattern scores do not generalise to real English.

4. Implementation and limitations

The Streamlit test bench supports model choice, temperature, seed, 10–500 samples, per-sentence tags and validity, four aggregate metrics, CSV export, a smoothed Viterbi tagger with an OOV warning, and a model/temperature comparison (screenshots on page 2). The generator can produce nonsense or reproduce training sentences; scores are relative to a synthetic corpus, not a measure of real language quality. Reproduction: pip install -r requirements.txt; python -m src.evaluate; python -m src.evaluate_real; streamlit run app.py. Dataset: Universal Dependencies English EWT, CC BY-SA 4.0. Source code: https://github.com/wijdane-ARB/03_chaotic_text_generator. Public app URL (Streamlit Community Cloud): *[to be added after deployment]*.

Application screenshots (local run)

Figure 1 – Generation: Plain HMM, T = 1.0, seed 42, 100 sentences.

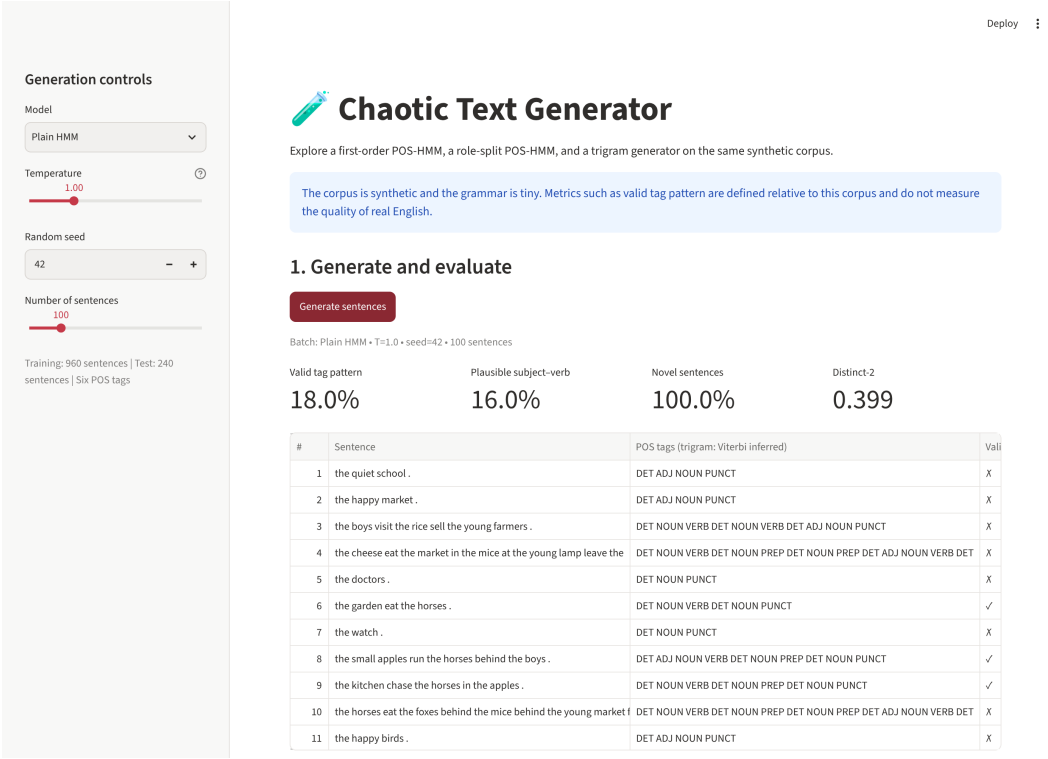


Figure 2 – Viterbi tagger with out-of-vocabulary warning (“zebras”).

